



DGS

DEVELOPMENT
GENERAL SYSTEM INDUSTRY
SRL

PRECISION IN EVERY PART. SPEED IN EVERY ORDER

Your strategic partner for premium multi-brand integration,
emergency custom manufacturing and guaranteed price stability

[FIND OUT MORE](#)

BUSINESS presentation & capabilities



DGSi

DEVELOPMENT
GENERAL SYSTEM INDUSTRY
SRL

About us

DGSi is a top-tier integrator and strategic supplier of **industrial solutions**, dedicated to optimizing the supply chain for companies in the automotive and manufacturing sectors. By combining a vast portfolio from global automation leaders (Siemens, Festo, Omron) with an exceptional capacity for the **rapid execution** of custom parts based on technical drawings, we reduce downtime and strengthen our clients' operations. We are committed to delivering not only technical excellence but also complete operational and financial predictability, backed by a firm **12-month price guarantee**.

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Why us

- **Premium Portfolio:** *Authorized distributor for dozens of global brands .*
- **Technical Agility:** *Emergency production for custom parts and welding electrodes based on technical drawings.*
- **Vendor Consolidation:** *Reducing time by acting as a Single Point of Contact for tools, automation, sensors and custom industrial parts.*
- **OPEX Predictability:** *Inflation shield with contractually locked prices for 1 full year.*

About company

Machining



- Holmaking & Boring
- Threading & Tapping
- Parting & Cutting-off
- Turning
- Milling

Toolholding



- Toolholders
- Collet Chucks
- Lathe Chucks
- Machine Vises
- Rigid clamping

Measurement tools



- Weight measurement
- Micrometers
- Feeler Gauges
- Dial indicator
- Calipers

Abrasives & Sawing



- Abrasive discs
- Wire brushes
- Saw blades
- Precision files
- Spare parts

Hand tools



- Assembly tools
- Torque tools
- Impact tools
- Cutting tools
- Tool sets



Audit & Technical Consulting

Production line & tooling analysis, "Stop-Line" risk identification, Strategic cost-down recommendations.



Sourcing & Procurement

1-Year Price Guarantee, "One Source" multi-brand portfolio, Budget & inventory optimization.



Emergency Manufacturing

Rapid intervention for line stoppages, Advanced ceramics, Custom-engineered welding electrodes.



Industrial Services

Tool Regrinding & Sharpening, Mold & Die Repair, Heat & Surface Treatments, Custom Part Manufacturing

Machining



Holemaking & Boring



Threading & Tapping



Parting & Cutting-off



Turning



Milling



Clamping



Toolholders



Drill Chucks



Lathe Chucks



Machine Vises



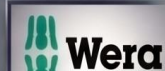
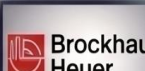
Rigid Clamping



PREMIUM MACHINING TECHNOLOGIES



PROFESSIONAL CLAMPING AND FIXTURING SOLUTIONS





Metrology :



Calipers



Micrometers



Dial Indicators



Feeler Gauges



Measurement Tools



Smart Sensors



Vision Systems



Laser Scanners



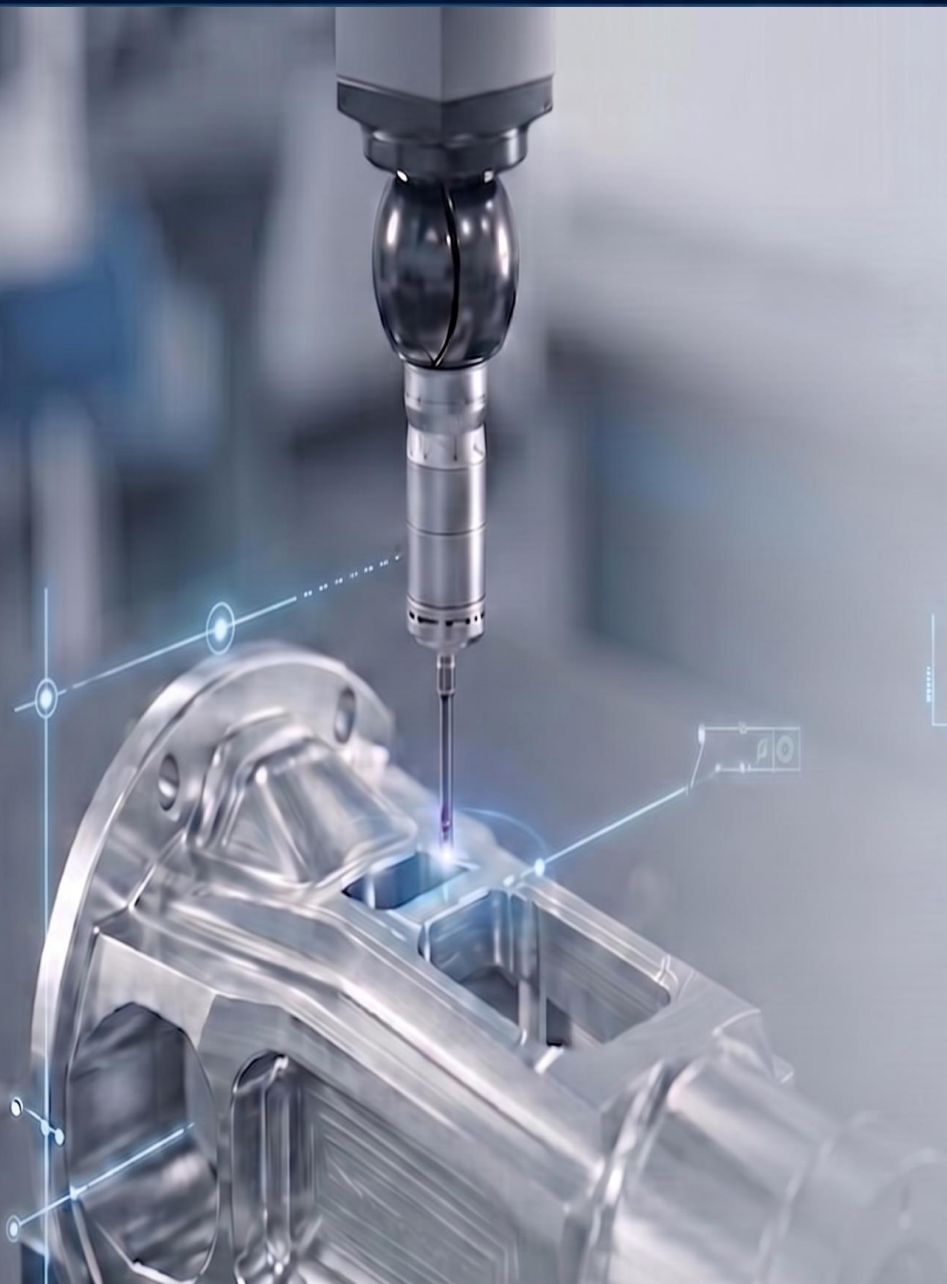
Proximity Sensors



Automation



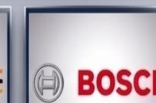
Industrial control



INDUSTRIAL METROLOGY



SENSORS





Hand Tools:



Fastening & Assembly



Torque Technology



Manual Machining



Hand tools



Tool sets



Protective Equipment



Hand protection



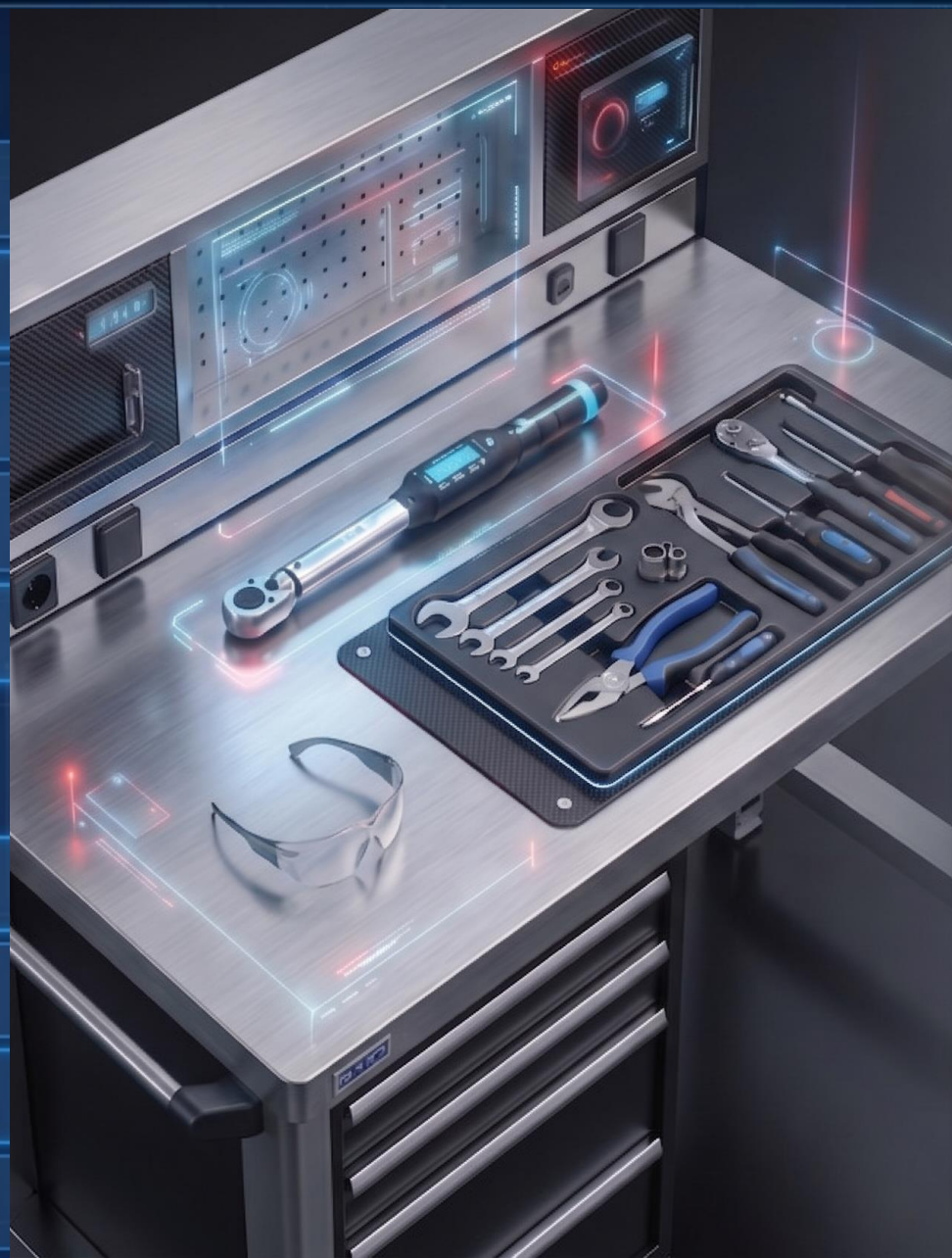
Head protection



Workwear & Footwear



Fall Protection & First Aid



PROFESSIONAL TOOLS



WORK EQUIPMENT





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Welding solutions

Extreme Performance through Advanced Ceramics & Modularity

- **Extended Lifespan** :Advanced ceramic components offer up to 20x greater durability than standard steel solutions, successfully withstanding extreme thermal and mechanical stress.
- **Perfect Insulation & Zero Defects**: Superior electrical and thermal insulation prevents short circuits and thread damage, ensuring consistent weld and significantly reducing scrap rates.
- **Optimized Maintenance (OEE / MTTR)**: Modular electrode architectures combined with rapid, custom manufacturing from technical drawings drastically reduce setup times and prevent unplanned line downtime.

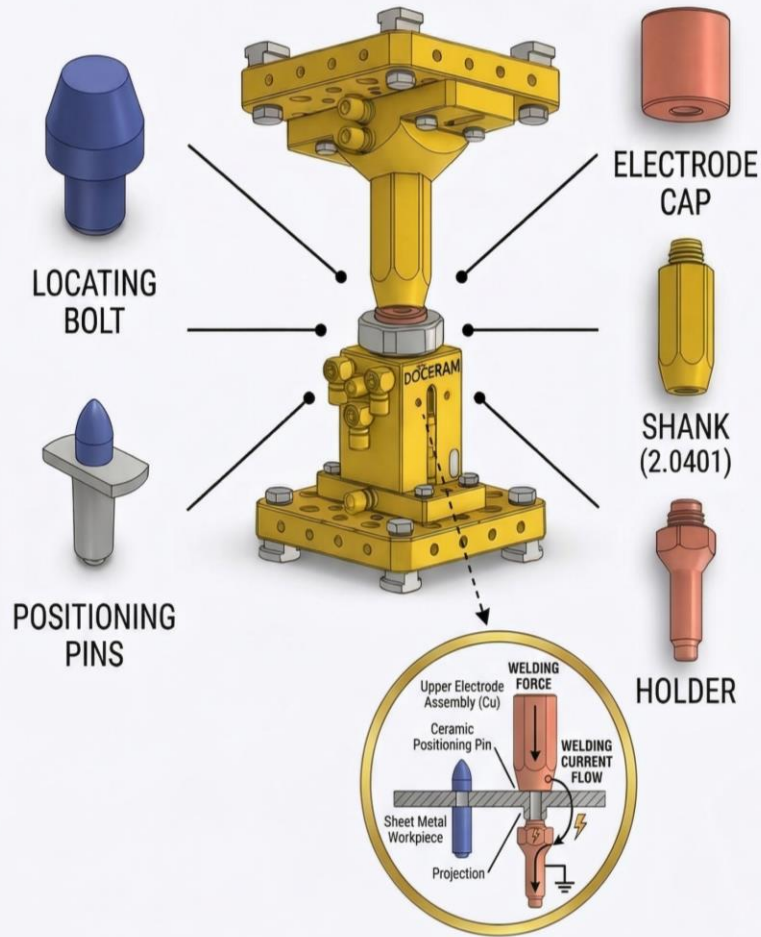
[FIND OUT MORE](#)

PERSONALISED solutions

Exploded view of Projection Welding

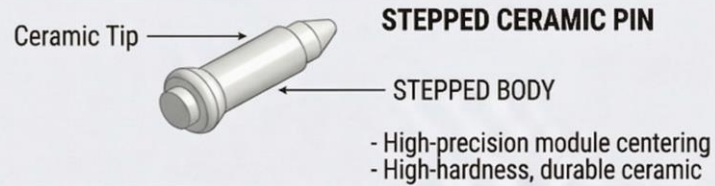
POSITIONING COMPONENTS (CERAMIC)

POWER ASSEMBLY (COPPER / ALLOY)

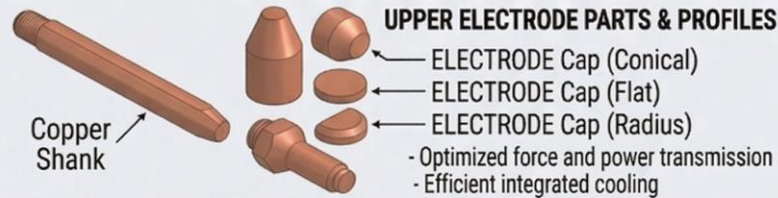


Components we deliver

UPPER CENTERING PINS



UPPER ELECTRODE PARTS



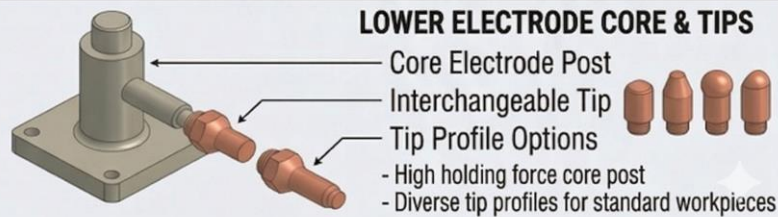
LOWER CENTERING PINS

INSULATED & PROTECTED PIN



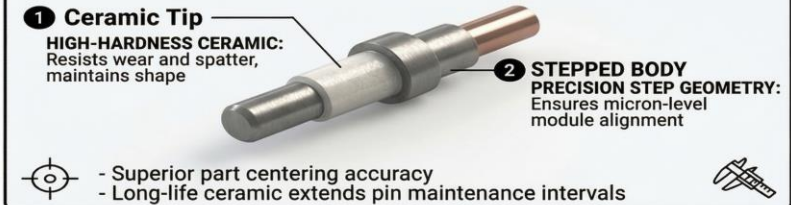
LOWER ELECTRODE TIPS

LOWER ELECTRODE CORE & TIPS



Key benefits we bring

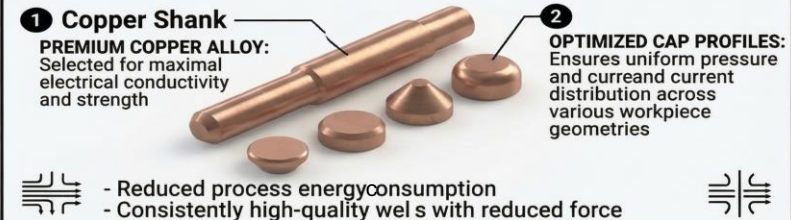
ADVANTAGE: PRECISION CENTERING



ADVANTAGE: ELECTRICAL ISOLATION



ADVANTAGE: OPTIMIZED POWER FLOW



ADVANTAGE: SYSTEM FLEXIBILITY



UPPER ELECTRODE MODULE

Key components. Our solutions

UPPER ELECTRODE: OPTIMIZED ENERGY TRANSFER

Core Function: Precise, consistent energy transfer for high-volume projection welding series.

Mechanical Rigidity: Eliminates deflection, ensuring stable power transmission and zero-defect joints.

Active Cooling: Directly manages thermal load at the welding tip.

ROI Impact: Extends wear electrode lifespan by up to **20x**, drastically reducing MTTR (Mean Time To Repair) and consumable costs.

Material Conformity:

- ❑ **Base/Holder/Shank/Tube:** 2.0401 (CuZn39Pb3) for maximum structural rigidity.
- ❑ **Wear Electrodes:** 2.1293 (CuCrZr) for premium thermal/electrical conductivity.

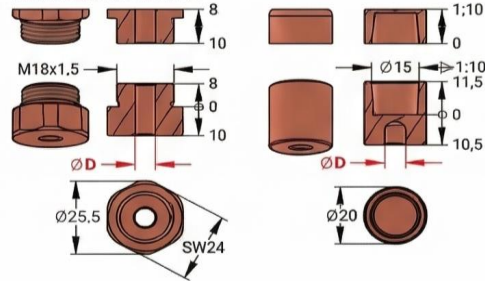
Dimensional Modularity:

- ❑ **Total System Length (L1):** 150 mm – 300 mm.
- ❑ **Shank Lengths (L):** 40 mm – 190 mm.
- ❑ **Tip Interfaces:** M18x1.5 (screw-in) or Cone 1:10 (cap).

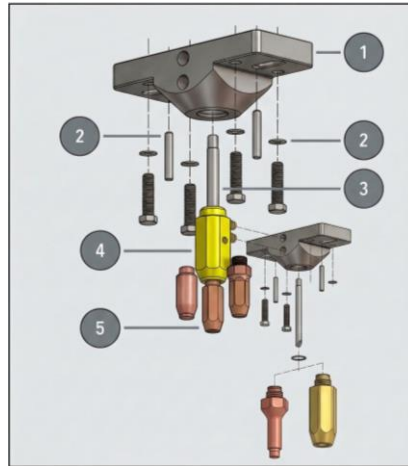
Cooling & Integration:

- ❑ **Fittings:** G1/4 connections (for Ø 8mm hoses) with NBR 70 O-rings.
- ❑ **Mounting:** Universal base plate compatible with 3-slot and 4-slot welding tables.

WEAR ELECTRODES



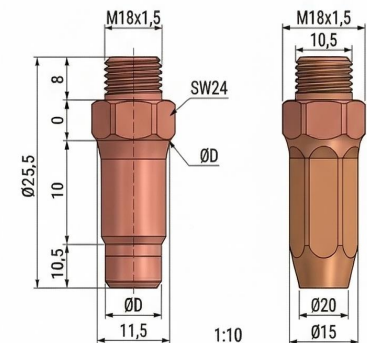
D [mm]	Screw-in Type	Cap Type	Example Code
3,0	Y	Y	111182418030
5,5	Y	Y	111181020050
7,0	Y	Y	111181020070
13,0	Y	N	111182418130



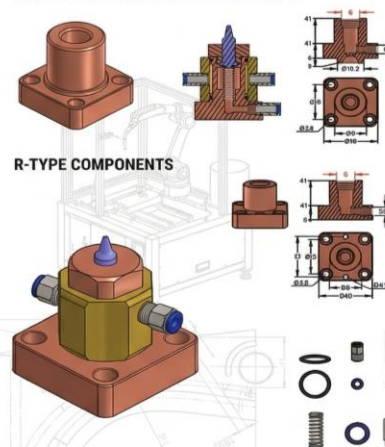
1: Electrode holder	4: Electrode shank
2: Accessories	5: Wear electrodes
3: Cooling tube	

Wear electrodes references and dimensions

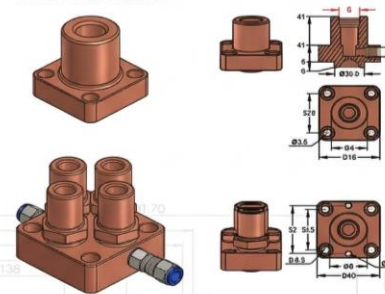
D	Screw-in electrode (2.1299)	Electrode cap (2.1293)
3,0	111182418030	111181020030
3,5	111182418035	111181020035
4,0	111182418040	111181020040
4,5	111182418045	111181020045
5,5	111182418055	111181020055
6,0	111182418060	111181020060
9,5	111182418065	111181020065
10,0	111182418070	111181020070



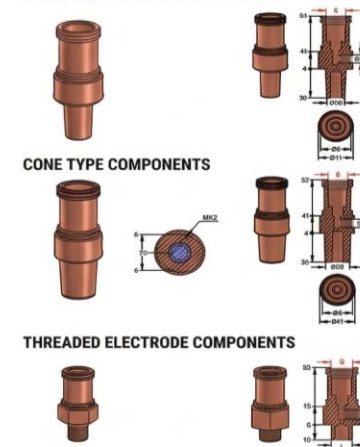
BASE ELECTRODES UNCOOLED



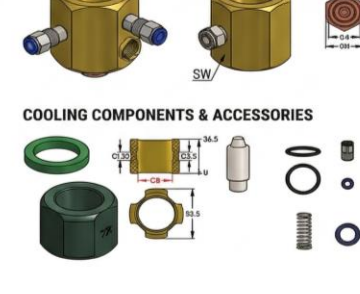
4XM6 COMPONENTS



BASE ELECTRODES COOLED



COOLING COMPONENTS & ACCESSORIES



Ceramic Pins key benefits:

- **Extreme Thermal & Shock Resistance:** Manufactured from high-performance ceramics like **Cerazur** or **Volcera**, these pins are designed to withstand intense welding temperatures and mechanical impact without deforming.
- **Precision Alignment for Zero-Defect Joints:** The wear-resistant nature of the ceramic ensures perfect, long-term positioning of components, which is critical for maintaining high repeatability and consistent weld quality.
- **Optimal Electrical Insulation:** As full-ceramic components, they provide essential insulation that prevents unwanted current shunting, ensuring that energy transfer is directed precisely through the electrode to the weld zone.

LOWER ELECTRODE MODULE

Key components. Our solutions

LOWER ELECTRODE: OPTIMIZED ENERGY TRANSFER

Core Function: Ensures consistent high welding quality and reliable component alignment, even under extreme conditions and in tight installation spaces.

Centering Precision: Utilizes full-ceramic centering pins to guarantee perfect, wear-resistant positioning and zero-defect joints.

Thermal Management: Available in active-cooled or uncooled solid-body configurations to withstand high temperatures and mechanical loads. ROI Impact: Exceptional material stability minimizes interruptions, sustainably reducing process and maintenance costs.

Material Conformity: Centering Pins: Cerazur or Volcera (high-performance full-ceramic) for extreme temperature and shock resistance. Exchange Electrodes: Premium copper alloys (e.g., 2.1285 / 2.1293) for optimal thermal and electrical conductivity. Insulation: DOGLAS for maximum compressive strength and thermal resistance.

Dimensional Modularity: Weld Nut Compatibility: Standard sizes from M4 to M16, and 7/16" UNF. Base Interfaces: Spigot (Ø18 mm – Ø30 mm), Thread M18x1.5, Cone 1:10, MK2, and MK3.

Cooling & Integration: Fittings: G1/8 connections (for Ø 6mm hoses) sealed with NBR 70 O-rings. Adaptability: Compact complete assemblies easily integrate into existing automated cells and tight installation spaces.

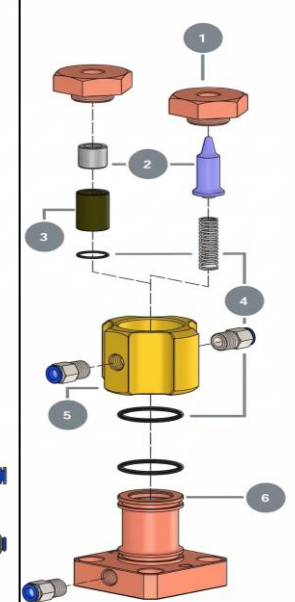
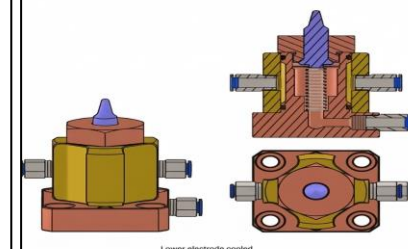
COMPLETE ELECTRODE

COOLED AND UNCOOLED
Complete electrodes offer a compact design that enables the production of precise welded joints, even in tight installation spaces.

Our cooled and uncooled complete electrodes are specially designed to withstand high temperatures and mechanical loads, while ensuring reliable centering through full-ceramic centering pins and sleeves.

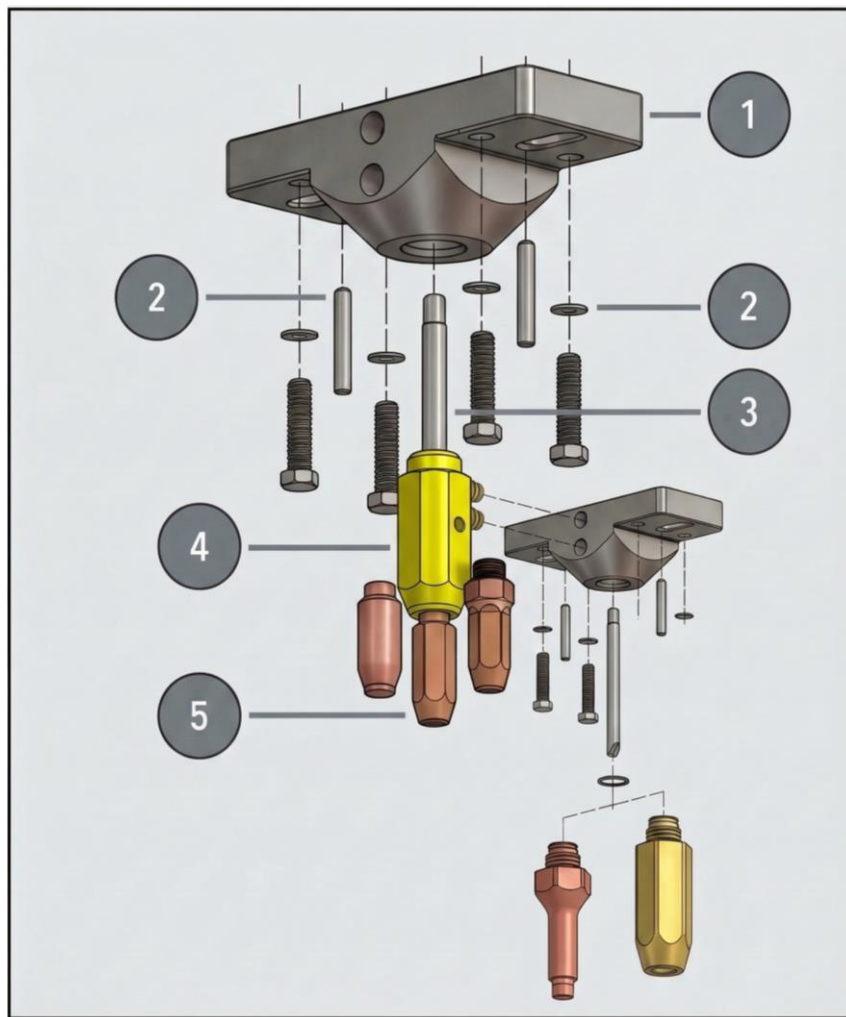
Their versatility and adaptability make them ideal for a variety of applications in industrial production.

Components at a glance:
1. Exchange electrode
2. Centering pin or sleeve made of Cerazur or Volcera
3. Insulating insert made of DOGLAS
4. Cooling ring of the cooled version
5. Accessories such as springs, compressed air connections and O-rings
6. Base electrode



UPPER ELECTRODE MODULE

Exploded view



1: Electrode holder	4: Electrode shank
2: Accessories	5: Wear electrodes
3: Cooling tube	

UPPER ELECTRODE: OPTIMIZED ENERGY TRANSFER

What it does:

Delivers the Energy: *It acts as the critical bridge that safely pushes massive amounts of electricity exactly where it needs to go to create a perfect, strong joint.*

Stays Cool Under Pressure: *It features an internal water-cooling system that fights off the extreme heat of the welding process, drastically extending the life of the machine.*

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Wear electrodes references and dimensions

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4,0	111182418040	111181020040
4,5	111182418045	111181020045
5,5	111182418055	111181020055
6,0	111182418060	111181020060
9,5	111182418065	111181020065
10,0	111182418070	111181020070

